

Evaluating the Base Excision Repair Inhibitor TRC102 and Temozolomide for patients with Recurrent Glioblastoma in the Phase 2 Adult Brain Tumor Consortium Trial BERT

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Supplementary Materials (attached as a separate files)

- **Supplementary Figure S1:** Supportive Preclinical Data
- **Supplementary Figure S2:** Signaling Pathway Impact Analysis of recurrent glioblastoma patients in the BERT trial.
- **Supplementary Figure S3:** Correlation between MGMT mRNA and MGMT protein abundance values in glioblastoma from the CPTAC database.
- **Supplementary Figure S4:** Correlation between MPG mRNA and MPG protein abundance values in glioblastoma from the CPTAC database.
- **Supplementary Figure S5:** Correlation between MGMT and MPG protein levels in glioblastoma from the CPTAC database.
- **Supplementary Figure S6:** Association between MGMT and MPG mRNA levels in newly-diagnosed glioblastoma patients from the TCGA database.
- **Supplementary Figure S7:** Association between MGMT and MPG mRNA levels in recurrent glioblastoma patients from the CCGA database.
- **Supplementary Table S1:** Key Resources Table
- **Supplementary Table S2:** Representativeness of Study Representativeness
- **Supplementary Table S3:** Adverse Events in the BERT trial
- **Supplementary Appendix S1:** Final Amended Version of the Clinical Trial Protocol

Abstract

Purpose: Patients with glioblastoma (GBM) have a dismal prognosis. While DNA alkylating agent temozolomide (TMZ) is mainstay of chemotherapy, therapeutic resistance develops rapidly in patients. Base excision repair inhibitor TRC102 (methoxyamine) reverses TMZ resistance in preclinical glioma models. We sought to investigate efficacy and safety of oral TRC102+TMZ for recurrent GBM (rGBM).

Patients and Methods: A pre-registered (NCT02395692), non-randomized, multicenter, phase 2 clinical trial (BERT) was planned and conducted through the Adult Brain Tumor Consortium (ABTC-1402). Arm 1 included bevacizumab-naïve GBM patients at first recurrence, with primary endpoint of response rates. If sufficient activity was identified, a second arm was planned in bevacizumab-refractory patients. Secondary endpoints were overall survival (OS), progression-free survival (PFS), PFS at six months (PFS-6), and toxicity.

Results: Arm 1 enrolled 19 patients with median of two treatment cycles. Objective responses were not observed, hence, arm 2 did not open. Median OS was 11.1 months (95%CI 8.2-17.9). Median PFS was 1.9 months (95%CI 1.8-3.7). PFS-6 was 10.5% (95%CI 1.3-33.1%). Most toxicities were Grade 1-2, with two Grade 3 lymphopenias and one Grade 4 thrombocytopenia. Two patients with PFS $\geq$ 17 months and OS >32 months were deemed ‘extended survivors’. RNA sequencing of tumor tissue, obtained at diagnosis, demonstrated significantly enriched signatures of DNA damage response (DDR), chromosomal instability (CIN70, CIN25), and cellular proliferation (PCNA25) in ‘extended survivors’.

Conclusions: These findings confirm safety and feasibility of TRC102+TMZ for rGBM patients. They also warrant further evaluation of combination therapy in biomarker-enriched trials enrolling GBM patients with baseline hyperactivated DDR pathways.

Keywords: Methoxyamine, TRC102, BER inhibitor, glioblastoma multiforme, high-grade glioma, N-methylpurine DNA glycosylase; MPG; TMZ

Translational Relevance:

In this multicenter, phase 2 trial conducted through the NCI Adult Brain Tumor Consortium in patients with recurrent glioblastoma (rGBM), DNA alkylating agent (temozolomide) combined with DNA base excision repair inhibitor (TRC102) was found to be safe and feasible for further evaluation in biomarker-enriched trials. A subset of patients benefitting from TRC102+TMZ was observed, in whom RNA sequencing of tumor tissue collected at diagnosis demonstrated significant enrichment for gene signatures of DNA damage response (DDR), chromosomal instability (CIN70, CIN25), and cellular proliferation (PCNA25). This DDR signature was found present in nearly 10% of all newly diagnosed glioblastoma patients in the Cancer Genome Atlas (TCGA). Findings of this trial demonstrate a potential for better patient selection for therapeutic targeting using a hyperactivated DDR gene signature. They also provide insights into the context of tumor vulnerability to synthetic lethality generated with the combination of a DNA damaging agent and DNA repair blockade.

Introduction

Glioblastoma (GBM) is the most common primary brain malignancy in adults.¹ It is also the most aggressive malignant brain tumor, with minimal therapeutic advances made in the last decade.² Despite standard of care therapy, 2- and 5-year overall survival (OS) rates remain 27% and 9.8%, respectively.³⁻⁵ while the median time to progression continues to be nearly seven months.³ Recurrence is nearly inevitable, for which no standard efficacious therapies currently exist. Salvage chemotherapies are largely ineffective, associated with median progression-free survival (PFS) at 6-months (PFS-6) rates of 10-15%.⁵ Bevacizumab, while being FDA-approved for recurrent GBM based on response rates, has not led to a significant increase in survival.^{6,7} Additionally, survival after the failure of bevacizumab administered in this setting is dismal, with a median survival of 4-6 months and PFS-6 of zero.⁸ Hence, novel therapies for recurrent GBM are urgently needed.

Temozolomide (TMZ), one of the rare pharmacological successes in the past few decades in GBM is the most commonly used chemotherapeutic agent, resulting in a survival benefit of 2-3 months.^{3,4} This second-generation alkylating agent causes DNA damage by attaching mutagenic adducts to tumor DNA leading to the generation of O⁶-methylguanine (O⁶-mG), N³-methyladenine (N³-mA) and N⁷-methylguanine (N⁷-mG) base lesions. The N³-mA and N⁷-mG adducts are repaired through base-excision repair (BER). The O⁶-mG adducts, the most cytotoxic lesions produced by TMZ, are repaired by mismatch repair (MMR) through O⁶-mG-DNA methyltransferase (MGMT).⁹ MMR's failure to fix O⁶-mG-thymine pairs results in long-lived DNA nicks and cell cycle inhibition. Thus, tumor sensitivity to TMZ is contingent upon low MGMT and MMR activity.

TMZ resistance is a critical element in GBM treatment failure.¹⁰ Resistance is secondary to increased rates of DNA repair, which occur either due to enhanced levels of repair proteins or defects in the MMR pathway.^{10,11} MGMT promoter demethylation permits increased MGMT expression followed by greater DNA repair, leading to rapid development of TMZ resistance and tumor growth, with recurrent tumors nearly always resistant to cytotoxic medication.^{12,13} MGMT methylation thus acts as a strong predictor of both TMZ response and OS.^{9,12,14,15}

However, O6-methylguanine lesions comprise ~5% of all TMZ-generated adducts, with the rest comprised of N3-methyladenine (~9%) and N7-methylguanine (>70%), which are acted upon by BER.^{16,17} Here, methyl groups in mutagenic adducts undergo cleavage of glycosidic bond by methylpurine DNA glycosylase (MPG), thereby generating apurinic/apyrimidinic (AP) sites.¹⁷ AP endonucleases (APE1/APEX1) then cleave the phosphodiester bond 5'- to the AP site. The resultant single-strand breaks (SSBs) are repaired by DNA polymerases through either non-displacing (short-patch) or displacing (long-patch) synthesis.¹⁸ Disruption of any step in this process results in BER dysfunction leaving adducts unrepaired and tumor cells at increased risk of apoptosis driven by double-strand breaks (DSBs).¹⁷

TRC102 (methoxyamine hydrochloride) is a small molecule BER inhibitor that binds to the sugar aldehyde at AP sites and prevents APEX1 from recognizing the damaged DNA site, thereby leaving the DNA unrepaired.¹⁹⁻²¹ TRC102's BER inhibition sensitizes both MMR-deficient and MMR-proficient tumor cells to methylating agents and thus potentiates TMZ cytotoxicity through induced SSBs and DSBs.¹⁹⁻²¹ Cancer cells treated with TMZ produce N⁷-mG and N³-mA DNA adducts, which activate BER to generate AP sites. TRC102 binds these sites covalently to form structurally modified AP sites resistant to APEX1's repair activity. This leads to targeting by topoisomerase II which induces double-strand breaks in DNA leading to tumor cell

apoptosis and illustrating a potential mechanism of synthetic lethality.²² This mechanism has been shown to reverse TMZ resistance in preclinical models.^{21,23} Work in human glioma cell lines has demonstrated that MPG overexpression greatly sensitizes tumor cells to temozolomide via BER inhibition through TRC102, but the latter is overturned by overexpression of BER-rate-limiting enzyme DNA polymerase β .²⁴ Higher MPG levels lead to increased binding sites for TRC102. MPG overexpression and BER blockade thus sensitize tumor cells to TMZ in a polymerase β -dependent fashion, indicating that MPG expression levels may predict TMZ potentiating activity of TRC102.²⁴ In CD133+ GBM stem cell xenograft mice models, TRC+TMZ significantly improved OS (median OS not reached after 230 days) versus TRC102 alone (79 days), saline controls (75 days, $P<0.001$), and TMZ alone (120 days, $P<0.01$) [Data from laboratory of Andrew Sloan, Case Comprehensive Cancer Center (CCCC), Cleveland, US; **Supplementary Figure S1**].

Given this strong preclinical rationale, the TRC102+TMZ combination therapy has been explored in initial trials. A phase I dose-escalation trial (NCT00892385) of IV TRC102 and oral TMZ in patients with advanced solid tumors and lymphomas, had indicated minimal toxicity and no DLTs of combination therapy, with the maximum tolerated dose (MTD) being 150 mg IV daily on day 1 and 200 mg/m² oral on days 1-5 for TRC102 and TMZ respectively, in combination therapy.²⁵ TRC102 pharmacokinetics were linear and unaffected by TMZ, with triallists recommending further investigation in tumors with demonstrated TMZ activity.²⁵ The present trial was therefore planned in rGBM patients, based on the significance of TMZ resistance in this patient population along with a significant unmet need. Additionally, gliomas have upregulated levels of APEX1, whose blockade leads to BER inhibition and greater sensitivity to TMZ.

Investigators in the National Cancer Institute (NCI) Adult Brain Tumor Consortium (ABTC) hypothesized that oral TRC102 can be safely co-administered with TMZ in patients with

rGBM and that TRC102 can augment DNA damage caused by TMZ, resulting in anti-rGBM responses. Therefore, a multicenter, phase 2 trial, named BERT, was planned and conducted by ABTC investigators to investigate the efficacy and safety of TRC102+TMZ therapy in patients with rGBM.

Methods

Study Design, Ethics, and Conduct

This study was a non-randomized, pre-registered (ClinicalTrials.gov Identifier: NCT02395692), multicenter, two-arm, two-stage Phase 2 efficacy trial, which was conducted in the US, and reported following the Consolidated Standards Of Reporting Trials (CONSORT) 2010 guidance.²⁶ The study was conducted through the National Cancer Institute (NCI) Adult Brain Tumor Consortium (Reference Code: ABTC-1402), whose predecessor was the New Approaches for Brain Tumor Therapy (NABTT) Consortium.^{27,28} The Key Resources Table for all associated data may be accessed in Supplementary Table S1. The trial protocol is available in the Supplementary Appendix S1.

The trial was conducted after approval by the Protocol Review Committee of the Cancer Therapy and Evaluation Program (CTEP Reference Number: PABTC_1402_R03PAPP01) and fulfillment of relevant local and national regulations, guidance, and policies, including the Declaration of Helsinki. The trial documents, including all amendments, were approved by the local Institutional Review Boards (IRBs) of all involved trial sites, including Johns Hopkins University (coordinating center), Cleveland Clinic, Ohio (IRB Reference number 15-1529), Henry Ford Health System (IRB Reference number 10163), University of Pittsburgh Medical Center (IRB Reference number PRO16020302; UPCI #15-171), University of California Los Angeles

(IRB Reference number 15-001680) and Wake Forest University Health Sciences (IRB Reference number IRB00034709). Written informed consent was provided prior to participation.

Study Objectives and Outcomes

Arm 1 included bevacizumab-naïve patients with the first recurrence of GBM following radiation therapy and TMZ. Arm 2 included bevacizumab-refractory GBM patients with an unlimited number of prior therapy regimens. The primary objective of arm 1 was to estimate the efficacy of oral TMZ+TRC102 as measured by overall objective response rate (ORR) according to the RANO criteria, with secondary endpoints being median PFS, PFS-6, OS, and toxicity.²⁹ Patients were evaluated at the end of every 2 treatment cycles (every 8 weeks). Assessment of response/progression was first performed at each institution by the local radiologist using RANO criteria. MRI scans of subjects showing tumor response were centrally reviewed by a neuroradiologist who independently assessed tumor sizes and computed percent tumor regression.

Arm 2 would open if arm 1 met ORR criteria, whose primary objective was evaluation of ORR in bevacizumab-refractory GBM patients, after administration of TRC102 and TMZ. The secondary endpoints of arm 2, similar to arm 1, were median PFS, PFS-6, OS, and toxicity.

The exploratory objective was to investigate the relationship between survival and biologic markers in tumor tissue, including MGMT, MPG, genomics, and transcriptomics, a design similar to the study of Weeks et al.³⁰ N-methylpurine DNA glycosylase (MPG) overexpression and inhibition of BER are associated with the sensitization of glioma cells to TMZ.²⁴

Study Participants

Patients with glioblastoma (GBM) who had progressed after surgical resection, external beam radiation, and adjuvant TMZ were included, with the representativeness of study participants further described in **Supplementary Table S2**. Further eligibility criteria included age ≥ 18 years,

with Karnofsky Performance Status (KPS) $\geq 60\%$ (i.e. be able to care for himself/herself with occasional help from others), adequate organ and marrow function defined as absolute neutrophil count $>1,500/\mu\text{L}$, platelets $>100,000/\mu\text{L}$, hemoglobin $>9\text{ g/dL}$, total bilirubin $\leq$ institutional upper limit of normal (iULN), AST/ALT ≤ 4 times iULN, creatinine $\leq$ iULN, creatinine clearance $>60\text{ ml/min/1.73m}^2$ for patients with creatinine levels above iULN, and APTT or PTT ≤ 1.5 times iULN. Patients were eligible for the trial if they were 12 weeks from the completion of radiotherapy and six weeks from nitrosourea chemotherapy. Additionally, while Response Assessment in Neuro-Oncology (RANO) criteria were not explicitly specified in patient eligibility, the trial was performed at established cancer centers with quaternary care brain tumor programs, where RANO criteria were used as a standard.

Patients with concurrent malignancy were excluded, except for curatively treated basal or squamous cell carcinoma of the skin or carcinoma in situ of the cervix, breast, or bladder, with the patient disease-free for ≥ 5 years. Other exclusion criteria were treatment with any other investigational agent, treatment with anticoagulants, active uncontrolled intercurrent illness, and/or prior gastrointestinal (GI) surgery or GI disease that might interfere with the absorption of TRC102. Pregnant subjects were also excluded from the study, given the potential teratogenic or abortifacient effect of TRC102. No exclusion to this study was based on race or ethnicity.

Study Intervention

TRC102 (methoxyamine) is a small molecule BER-inhibitor developed by investigators from CCCC (Cleveland, Ohio, US) and an oral formulation was supplied by NCI Division of Cancer Treatment and Diagnosis (DCTD, RRID:SCR_004196), under a collaborative agreement with Tacon Pharmaceuticals, Inc. (San Diego, CA).

TRC102 and TMZ were administered orally, based on prior clinical trials. While the trial at CCCC (NCT00892385) had dosed TRC102 intravenously,²⁵ the NCI phase I trial in relapsed solid tumors and lymphomas had dosed TRC102 orally on days 1-5 of a 28-day cycle (NCT01851369), along with oral TMZ.³¹ The phase 1 trial had determined the maximum tolerated dose (MTD) as 150 mg TRC102 and 150 mg/m² TMZ. Based on these findings, dosing in the current trial was planned, where TMZ was dosed based on body surface area at 150 mg/m²/day, while TRC102 was dosed at flat 150mg/day, both administered orally once a day for 5 consecutive days, repeated every 28 days, with weekly evaluation of blood counts. No dose modifications were required.

Enrollment was done using the Oncology Patient Enrollment Network (OPEN) and supported by the NCI Cancer Trials Support Unit (CTSU). Treatment was administered on an outpatient basis. Patients were allowed to receive other medications that the investigator deemed to be medically necessary, with the specific exception of non-protocol-specified chemotherapy, radiotherapy, anti-neoplastic biological therapy, or other investigational agents.

Molecular Testing

Tumor MGMT status was obtained at individual treatment sites. The results of routinely used methods for routine MGMT promoter methylation testing (for instance, methylation-specific PCR or quantitative PCR), as per institutional standards, were considered acceptable. Gene expression analyses were performed by the Translational Genomics Research Institute (TGen, City of Hope, CA, RRID:SCR_011581). Immunohistochemistry for MPG expression was performed on pretreatment paraffin-embedded sections from representative areas of tumors using rabbit polyclonal anti-MPG antibody (Sigma-Aldrich Cat# HPA006531, RRID: AB_1854075) at 1:800 dilution with citrate buffer antigen retrieval MACH 4 HRP Polymer (BioCare Medical, Pacheco,

CA) and Betazoid DABKit (BioCare Medical, Pacheco, CA) detection method. MPG expression was graded from 0 to 2+. Slides were reviewed in a blinded fashion by a neuropathologist. RNA sequencing was performed, under a separate IRB restricted to specific trial sites, on tumor tissue from treatment-naïve recurrent GBM patients. From the samples received for genomic analyses at a secondary site (TGen, City of Hope, CA), sequencing was performed on samples with appropriate RNA integrity numbers in order to understand the context of vulnerability to TRC102 and TMZ in recurrent GBM patients.

Analysis of Clinical Data

The primary objective was to test the hypothesis of whether the combination therapy of TRC and TMZ would achieve a 30% overall radiographic response rate (partial response and complete response) of TRC102+TMZ in patients with the first recurrence of GBM, based on criteria set by NCI-CTEP protocol review committee.

A two-stage (mini-max) design was used to discriminate between true response rates of no more than 13% and at least 30% with 85% statistical power and a 10% false positive rate in arm 1. The maximum trial size would be set at 31 evaluable patients. If at least seven responses (at least 22.5%) were observed among the 31 evaluable patients, this regimen would be considered worthy of further testing in this disease. If no more than two responses (no more than 10.5%) were observed among the initial 19 patients, the study would be terminated early for futility.

Arm 2 for bevacizumab-refractory patients was warranted only if the first arm met its efficacy criteria. A two-stage design for Arm 2 with 90% statistical power and 5% false positive rate to test the hypothesis of a clinical benefit of 30% radiographic response against a 10% overall

response rate of TRC102+TMZ in patients with multiple prior therapies and were bevacizumab-refractory, treated with TRC102 and TMZ, and continued on bevacizumab.

Nineteen GBM patients at first recurrence were enrolled and included in the analysis (arm 1 and stage 1 only). The time of progression-free survival was calculated from the date treatment started to the date of disease progression or censored during analysis if the patient was free of progressive disease. The two longest surviving patients received extended follow-up after the dataset lock. OS duration was calculated from the start date of treatment to the date of death or censoring at the time of analysis if patients were alive. Baseline patient characteristics were summarized using standard descriptive summaries. The PFS-6 proportion was estimated using binomial distribution with a 95% confidence interval. OS and PFS were estimated using the Kaplan–Meier method, with confidence intervals constructed using the method of Brookmeyer-Crowley.³² All adverse events were graded according to Common Terminology Criteria for Adverse Events (CTCAE) version 4.0. Adverse events deemed to be possibly, probably, or definitely attributed to TRC102+TMZ were summarized using descriptive statistics. Log-rank test was used to explore the survival differences between GBM patients with MGMT-methylated and unmethylated tumors. All statistical analyses were performed using SAS version 9.4 (SAS Institute, Cary, NC, RRID:SCR_008567).

Analysis of Molecular Data

Clinical outcomes were correlated with tumor molecular features, including MGMT and MPG status, and reported in accordance with elements of the REMARK reporting framework.³³ After RNA sequencing of tumor tissue obtained at baseline, functional class scoring was pursued using Gene Set Enrichment Analysis (GSEA) and single sample GSEA. For this, differentially

expressed genes (DEGs) were analyzed using DESeq2 GSEA (RRID:SCR_015687; RRID:SCR_003199). This was applied to transform the expression profiles into enrichment scores for the 'HALLMARK_DNA_REPAIR' gene set (44 founder gene sets) from the Molecular Signatures Database (MSigDB, RRID:SCR_016863), using methods we have described previously.³⁴ GSEA and clustering analysis were used to explore and assign candidate mechanistic signatures of distinct vulnerability to TRC102+TMZ.

We also evaluated patients against a set of 39 manually curated gene sets to understand if these gene sets could also differentiate responder and non-responder patients. These gene sets, in MSigDB, were associated with "DNA Repair", "Base Excision Repair", "Nucleotide Excision Repair", TP53 signaling (given its role in driving BER signaling), tumor mutational burden for GBM, and chromosomal aberrations, given the association between genomic instability and DNA damage response.

Single sample GSEA was performed in R using Bioconductor package (RRID:SCR_006442) using methods described by Gu et al.³⁵ Signaling Pathway Impact Analysis (SPIA) integrating differentially expressed genes and their log fold changes with signaling pathways topology was performed as described by Tarca and colleagues.³⁶ Comparative analyses were performed with datasets from the Clinical Proteomic Tumor Analysis Consortium (CPTAC, RRID:SCR_017135), The Cancer Genome Atlas (TCGA, RRID:SCR_003193), and the Circulating Cell-free Genome Atlas study (CCGA, NCT02889978).

Data Availability

The final amended study protocol is available in Supplementary Appendix S13. This study, a clinical trial, generated individual patient data (IPD), which were used to create the

figures and the tables in the main manuscript. Additional analyses carried out on publicly available datasets are reported in the supplementary figures S3-S7. Deidentified IPD is not being made available publicly given the small sample size (N=19), the relatively low incidence of glioblastoma (compared to other systemic malignancies), and the novelty of the study medication, all of which together place limits on assurance of patient privacy and confidentiality. IPD data, including source data for figures and tables, informed consent form, and clinical study report, will be made available to qualified researchers for IPD meta-analysis upon reasonable request for up to six years after publication. Proposals should be directed to the corresponding author at manmeetA@baptisthealth.net.

Results

The trial began patient enrollment in January 2016 and finished enrolling in June 2016. Nineteen patients were enrolled in arm 1 in the first stage of the trial, i.e., TRC102+TMZ for bevacizumab-naïve recurrent GBM (rGBM). The median age of patients was 60 years (range 48-76). The median KPS at baseline was 80 (range 70-90). 53% of the patients were female, and 95% were white. The median number of treatment cycles was 2 (range 1-12). The baseline patient attributes and the disease characteristics are summarized in **Table 1** and compared with prior literature in **Supplementary Table S2**. No dose modification was performed for any recruited patient.

Clinical Outcomes

Among the 19 patients included in the analysis, 17 had died by the time of the dataset lock. The study did not meet its prespecified response rate criteria and was terminated after arm 1/stage 1. Hence, arm 2 did not open for accrual. At the time of the planned database lock, the median OS

was 11.1 months (95%CI 8.2-17.9 months) and the median PFS was 1.9 months (95%CI 1.8-3.7 months) (**Figure 1**). PFS at six months was 10.5% (N=2/19, 95%CI: 1.3-33.1%). Five patients were found to have an OS of ≥ 20 months. The combination of TRC102 and TMZ was safe and well tolerated. Overall AEs that were possibly, probably, or definitely related to either TRC102, TMZ, or both are listed in **Supplementary Table S3**. Most treatment-related AEs were CTCAE Grade 1 or 2, with only two CTCAE Grade 3 lymphopenias and one CTCAE Grade 4 thrombocytopenia.

There were two patients deemed ‘extended survivors’, where extended follow-up beyond the dataset lock date indicated a PFS of ≥ 17 months and OS ≥ 32 months. Both were aged >55 years, had KPS of 80, and had IDH wild-type and MGMT promoter methylated tumors. Following treatment in the trial, one patient had a PFS of 29 months and was alive at 60 months when lost to follow-up, while the other had a PFS of 17 months and an OS of 32 months.

Correlation with MPG and MGMT expression

MGMT promoter methylation status was available for all included patients. 14/19 patients had MGMT unmethylated tumors. The median OS of methylated patients was 30.2 months (95%CI 9.5-NA), and in unmethylated patients was 9.9 months (95%CI 5.8-13) (P=0.02). Median PFS in methylated patients was 5.6 months (95%CI 1.9-NA) and in unmethylated patients was 1.9 months (95%CI 1.8-2.3) (P=0.003).

Tissue for MPG expression was available in 14 cases. Nuclear positivity for MPG via IHC was identified in eight patients (1+ in five, 2+ in three). MPG expression was negative in six. Stratified analysis of PFS and OS using MPG expression status demonstrated an inverse relationship. Median OS was 11.1 months (95%CI 3.7-NA, N=3) for 2+ MPG expression, 14.5

months (95%CI 7.9-32.3) for 1+ MPG expression and 17.9 months (95%CI 9.3-30.2) for absent expression (P=0.05). Median PFS was 1.9 months (95%CI 1.4-NA) for 2+ MPG expression, 2.8 months (95%CI 1.9-NA) for 1+ MPG expression, and 3.7 months (95%CI 1.8-5.6) for absent expression (P=0.07). In the 5 patients where MPG expression could not be evaluated due to trial logistics, the median OS was 8.1 months (95%CI 2.0-10.2), while the median PFS was 1.8 months (1.6-1.9). However, one ‘extended survivor’ had 1+ MPG expression while the other had 2+ MPG expression.

Translational Findings

Adequate quantity and quality of RNA samples were available for transcriptome profiling from the tumor tissue of 7 patients. In the two patients deemed ‘extended survivors’ (PFS $\geq$ 17 months), preliminary analysis on RNA sequencing data revealed significant enrichment for DNA damage response (DDR) pathways (MSigDB) (**Figure 2A**), chromosomal instability gene signatures (CIN70 and CIN25) (**Figure 2B**), and aneuploidy that includes proliferative gene signature (PCNA25) (**Figure 2C**). Another patient with OS of 30.2 months and PFS of 5.5 months (patient ID 0205, IDH wild type, MGMT methylated) had some borderline expression for DDR. Extended disease control with TRC102 + TMZ combination therapy was found independent of the molecular subtype of glioblastoma (mesenchymal, classical, or proneural) (**Figure 3A**). Single sample GSEA with hierarchical clustering of thirteen genesets segregated these ‘extended survivors’ from those whose disease progressed early (**Figure 3B**). Genesets encompassing nucleotide excision repair featured prominently.

Analysis of 178 GBM patients (RNASeq) in the Cancer Genome Atlas (TCGA) database using 44 founder gene sets in HALLMARK_DNA_REPAIR (MSigDB) also demonstrates that

9.6% (N=17) of glioblastoma patients in TCGA have hyperactivation of DNA Damage Response signaling pathways (**Figure 3**).

Focusing on BER-specific gene sets, which included MPG expression and 18 other genes culled in the REACTOME (**Figure 4A**) or 38 other genes culled in KEGG (**Figure 4B**), did not clearly segregate the extended survivors from other patients. Meanwhile, cluster analysis using 39 manually selected genesets, including DNA Repair, Base Excision Repair, and Nucleotide Excision Repair, was performed, which segregated extended survivors from those with worse disease course (**Figure 4C**).

Signaling Pathway Impact Analysis (SPIA) using differentially expressed genes and their log fold changes integrated with signaling pathways topology uncovered pathways differentiating extended survivors from others (**Supplementary Figure S2**). Notable pathways inhibited in ‘extended survivors’ include KEGG 5200 “Pathways in Cancer” (313 genes), KEGG 4810 “Regulation of actin cytoskeleton” (204 genes), KEGG 5218 “Melanoma” (68 genes), and KEGG 4540 “Gap junction” (88 genes).

Comparative analyses done using cBioPortal, the TCGA database, the CCGA database, and the CPTAC database provided additional insights into the interplay between MGMT and MPG status (**Supplementary Figures S3 to S7**). A substantially high correlation was found between mRNA Z-scores and protein abundance scores (Spearman’s coefficient 0.79, $P<0.001$) for MGMT but not for MPG (Spearman’s coefficient 0.15, $P=0.15$) in the CPTAC database (**Supplementary Figures S3 and S4**). The protein abundance scores of MGMT and MPG did not correlate well in the CPTAC database (Spearman’s coefficient -0.10, $P=0.317$) (**Supplementary Appendix S5**). Meanwhile, mRNA co-expression values for MGMT and MPG were found to be well correlated for primary GBM in the TCGA database (Spearman’s coefficient 0.51, $P<0.001$) (**Supplementary**

Appendix S6), and for recurrent GBM in the CCGA database (Spearman's coefficient 0.58, $P < 0.001$) (**Supplementary Appendix S7**).

Discussion

The past three decades have seen only a single efficacious drug being FDA-approved for glioblastoma, i.e., temozolomide, with the vast majority of neuro-oncology clinical trials failing to translate investigational therapies to the standard of care. Unlike other solid tumors, efficacious drugs still remain unavailable after glioblastoma (GBM) recurrence, despite the thousands of patients accrued to neuro-oncology trials worldwide.³⁷ Therefore, careful investigation into novel therapeutics remain vital to improving outcomes for patients with progressive GBM.^{37,38} In this work, we reported findings from a multicenter phase 2 clinical trial (BERT), along with correlative transcriptomic analyses, of an investigational DNA base excision repair (BER) inhibitor TRC102, combined with the DNA alkylating agent temozolomide for patients with bevacizumab-naïve recurrent GBM at first relapse. The results of the trial confirm that TRC102, in combination with TMZ, is safe, well-tolerated, and therapeutically feasible for patients with recurrent GBM. Trial findings also provide insights into the context of tumor vulnerability to synthetic lethality generated with a combination of a DNA damaging agent and DNA repair blockade, along with indicating a potential for improved patient selection for therapeutic targeting through DNA damage response (DDR) signature.

Given the extremely low rates of successful bench-to-bedside drug translations for glioblastoma, optimizing the currently available treatments is a meaningful approach. TMZ resistance is associated with the expression levels of DNA alkylating proteins and DNA repair enzymes.³⁹ Significant efforts have been directed at sensitizing resistant tumors to currently available cytotoxic therapies such as TMZ. Because the BER pathway is a principal cellular

process employed by malignant cells to reduce the cytotoxicity of alkylating agents, the combination of TMZ with a BER inhibitor has been hypothesized to improve anti-tumor efficacy, which was demonstrated preclinically (**Supplementary Figure S1**). This phase 2 trial depicts the safety and feasibility of a first-in-class inhibitor of BER TRC102 combined with TMZ for patients with GBM progressive after first-line therapy with TMZ and radiation therapy. Since the pre-specified criteria were not met in the present study, the trial did not proceed to test TRC102 and TMZ in bevacizumab-refractory tumors, a particularly treatment-refractory subset of patients where all efforts to induce relevant disease stabilization have thus far failed. This failure to demonstrate clinical efficacy may be seen in the background of a progressive disease where the vast majority of trials fail to meet their primary endpoint. Here, the trial had >70% of patients with MGMT unmethylated tumors, who were unlikely to respond to TMZ.⁴⁰ While efficacy here was limited, the combination of TRC102 and TMZ may hold further potential in a biomarker-enriched subset of patients. In the CCCC trial, Eads et al. had reported a lack of correlation between dose level and either COMET assay outcome or response to therapy.⁴¹ However, their pharmacodynamic assays had been performed on peripheral blood mononuclear cells, rather than tumor tissue, which the investigators proposed assessing through quantitative assessment of AP sites in tumor tissue.³⁹ In the current trial, a therapeutic vulnerability signature was detected by sequencing tumor tissue obtained at initial diagnosis, i.e. prior to TRC102+TMZ therapy.

Here, TRC102 did not significantly potentiate TMZ-induced myelosuppression, and non-hematologic toxicity was rare. This is consistent with the prior Case Comprehensive Cancer Center (CCCC) trial,²⁵ and the NCI phase 1 trial,³¹ both trial reports published in full recently.^{41,42} The CCCC trial had a mixed patient population including 29% colorectal, 16% lung, 13% pancreas, 8% head and neck, and 5% neuroendocrine cancers amongst others. Anemia occurred in half of

all enrolled patients, being majorly grade 1-2 and not associated with hemolysis. Eads et al. had also reported a single occurrence of grade 4 thrombocytopenia amongst 38 patients,⁴¹ similar to the current trial with one grade 4 thrombocytopenia. Meanwhile, of the 52 patients enrolled in the NCI trial, 30% had colorectal cancer, 12% had ovarian cancer, 8% had mesothelioma, 6% had breast cancer, and 6% had cholangiocarcinoma, among others.⁴² Here as well, most toxicities were grade 1-2, with grade 3 or higher anemia hematologic toxicities including anemia in 10 patients, lymphopenia in 6 patients, and thrombocytopenia in 3 patients (all grade 4).

Patients in this trial having MGMT methylated tumors had improved survival, consistent with the prior literature on glioblastoma on the strong prognostic utility of MGMT silencing, the critical role of DNA repair, and the utility of targeting DDR.^{40,43-44} Meanwhile, upon stratification by MPG status, the median survival was longest in those lacking MPG expression, shorter in the 1+ expression group, and shortest in the 2+ expression cohort. Considerable literature exists on the conflicting results regarding the role of MPG in tumor cell vulnerability to TMZ and the prognostic significance of MPG expression in malignant gliomas. Fosmark et al. found a significant association between high levels of MPG and improved OS,⁴⁵ consistent with the results from both the TCGA dataset and *in-vitro* studies.^{24,46} However, this contrasts with other studies showing that patients with MPG non-expression had a more favorable outcome.^{47,48} The latter is supported by data from studies in patient-derived glioma cells, where high MPG expression correlated with a TMZ-resistant phenotype.¹⁸ Confounding a simple direct mechanistic link between low MPG and greater vulnerability to the genotoxic effects of TMZ, another report by Chou et al. in 2015 indicated lethal DDR activation by base lesions that trigger the ATM-Chk2 pathway.⁴⁹ We found that while levels of MGMT and MPG mRNA are concordant for both primary and recurrent glioblastoma (**Supplementary Figures S6 and S7**), this correlation fails to materialize at the level

of proteins, where MPG mRNA values do not correspond to MPG protein values (**Supplementary Figure S4**), nor do MPG protein levels correspond to MGMT protein levels (**Supplementary Figure S5**). This potentially supports the use of immunohistochemistry (IHC) for the detection of MPG status in GBM, as performed here, instead of RNA sequencing. This contrasts with MGMT, where protein abundance correlates well with mRNA expression (**Supplementary Figure S4**). Overall, given the discordant results in our trial and as demonstrated in prior studies, MPG remains a challenging biomarker for investigation, with likely several secondary mechanisms at play.^{18,24,45-}

⁴⁸

In this trial, analysis of transcriptomic genesets of BER alone (which include MPG expression) did not segregate the extended survivors from other patients (**Figure 4B**). The existence of multiple DNA-repair-related pathways potentially obscures the fine segregation of extended survivors and also makes it challenging to interpret to what degree the differential expression of each specific pathway contributes to the likelihood of therapeutic response. While TRC102 inhibits the BER pathway, there may be off-target effects that account for *in vivo* antiglioma activity. Further, the work by Chou et al, describing the lethal DDR activation by DNA lesions triggering the ATM-Chk2 pathway, ties into the more recent work by Dmello and colleagues implicating Chk2 expression by glioma cells with escape from recognition by CD8 T-cells.^{49,50} BER inhibition in some glioblastoma patients may potentially have sensitization consequences to immune cell attack of the tumor. Such off-target consequences as immune sensitization in the clinical setting may be a source of the discordance related to biomarkers of BER inhibition, including MPG protein levels and BER-specific genesets.⁵⁰

In this trial, two ‘extended survivors’ demonstrated substantial evidence of clinical benefit. Not only did they meet the secondary endpoint of PFS at six months, but they also had considerably

longer duration of survival (32 months and 60 months) and durable disease control, as evident in in PFS of ≥ 17 months. Both had MGMT promoter methylation which also contributed to reduced levels of DNA repair in tumor cells. RNA sequencing of their tumor tissue revealed significant enrichment for DNA damage response pathways, chromosomal instability gene signature, and proliferative gene signature. These molecular signatures could serve as useful biomarkers to identify GBM patients likely to respond to a DNA alkylating agent combined with a DNA repair inhibitor.

Prospective clinical validation of a molecular signature that predicts vulnerability to DNA repair inhibitors in malignant gliomas will have applications in other malignancies with resistance to DNA damaging agents. The trial investigators are planning a biomarker-enriched clinical trial where prospective validation of the DDR signature will be pursued. This proposed phase I/II trial is planned to enroll newly diagnosed, MGMT methylated GBM patients with hyperactivated DDR molecular signature at baseline. Post-resection, TRC102 will be administered, in a 3+3 dose-escalation design, concurrently with temozolomide and focal radiation therapy. It is hypothesized that these patients with hyperactivated DDR will potentially respond better to combinatorial therapy of TRC102 and TMZ, both of which impact DNA repair pathways.

Finally, the use of a glioblastoma umbrella signature trial (GUST), which tests multiple investigational arms based on optimal patient selection using biochemical and molecular attributes, may represent a suitable strategy for employing a host of molecular signatures to stratify patients into various arms based on their likelihood of response.^{28,37,51} We have recently utilized semi-supervised machine learning approaches on the TCGA database to identify signatures of vulnerability for GBM. Using Entropy-Regularized Logistic Regression (ERLR), a predictive model was developed based on both genomics and transcriptomics, transforming the unclassified

TCGA GBM patients into self-labeled datasets with the highest likelihood of response.⁵² This permitted a virtual/mock stratification of >50% of the dataset's patients into at least one of four treatment arms, having an elevated likelihood of response to TRC102 (DNA BER inhibitor), arsenic trioxide (pleiotropic action), selinexor (exportin-1 inhibitor),⁵³ or pevonedioistat (neddylation inhibitor).⁵⁴ Use of innovative trial designs will likely be critical to success in the extremely challenging space of GBM.^{28,37,51}

Limitations of the Study

First, given the need to demonstrate initial utility in a phase 2 trial, logistical and funding requirements precluded the pursuit of a randomized controlled investigation. Second, the trial's second arm did not open for accrual as the first arm did not meet prespecified criteria, thus limiting definitive conclusions related to the efficacy of this novel medication. Third, a lack of data on several molecular markers of GBM, now considered well-established as part of routine care, could not be fully collected as several ABTC investigators changed institutions. Despite best efforts, challenges related to tumor tissue access precluded the tallying of patient data regarding these genetic markers from some of these institutions. Fourth, the transcriptomic analysis of tumors was done on tissue collected at initial diagnosis, which may be of mixed utility in depicting the transcriptome of these tumors at the time of recurrence when these patients were enrolled in the clinical trial. However, since the pre-therapy transcriptomic data was available, it did permit the prospective identification of the subset of patients with hyperactivated DNA damage response who would potentially benefit from TRC102+TMZ combination therapy. Finally, the representativeness of tumor tissue transcriptomics is confounded by the complexity of the full complement of host cells along with tumor cells in resected tissue as well as the widely appreciated heterogeneity of GBM.

Conclusions

DNA base excision repair inhibitor TRC102, in combination with DNA damaging agent TMZ, is safe and well-tolerated in patients with recurrent or progressive GBM. A hyperactivated DNA damage response (DDR) molecular signature at baseline in GBM patients is associated with extended disease control on-therapy with TRC102 and TMZ, which needs prospective validation in a future clinical trial.

Abbreviations

ABTC	Adult Brain Tumor Consortium
BER	Base Excision Repair
DDR	DNA Damage Response
GBM	Glioblastoma
MGMT	Methyl-Guanyl Methyl-Transferase
MTD	Maximum Tolerated Dose
NCI	National Cancer Institute
OS	Overall Survival
PFS	Progression-Free Survival
TMZ	Temozolomide

DECLARATIONS:

Ethics approval and consent to participate: The trial was conducted in accordance with all relevant regulations and guidelines, including the declaration of Helsinki. Ethics approval was provided by the local Institutional Review Boards (IRBs) of all involved trial sites, including Johns Hopkins University (coordinating center), Cleveland Clinic, Ohio (IRB Reference number 15-1529), Henry Ford Health System (IRB Reference number 10163), University of Pittsburgh Medical Center (IRB Reference number PRO16020302; UPCI #15-171), University of California Los Angeles (IRB Reference number 15-001680) and Wake Forest University Health Sciences (IRB Reference number IRB00034709). Written informed consent for participation was obtained from all individual participants included in the study.

Consent for publication: Consent for publication of the findings of the trial was obtained.

Inclusion and Diversity: We support inclusive, diverse, and equitable conduct of research.

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623

Competing Interests:

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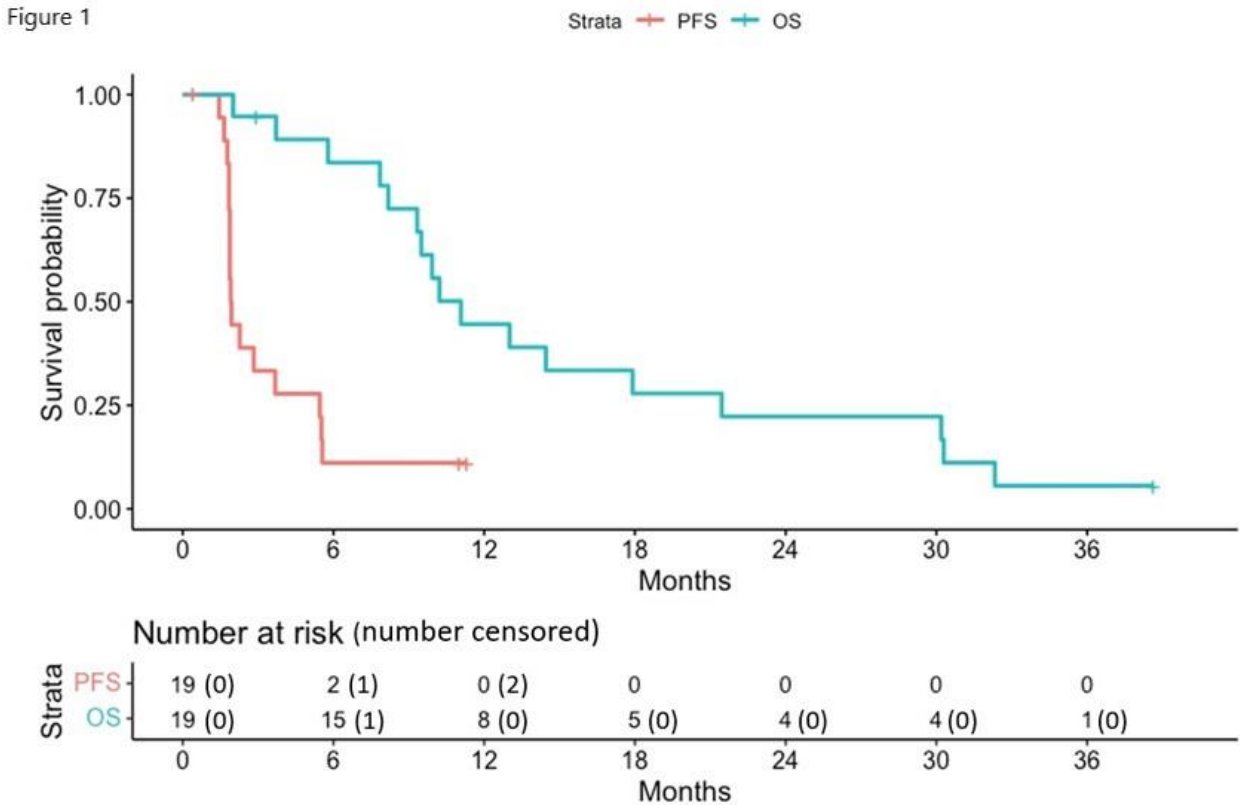
All other authors have no relevant disclosures.

Main Manuscript Tables and Figures:**Table 1:** Baseline patient characteristics at enrollment in the ABTC-1402 Phase 2 Trial.

Characteristics	Values
Patients Enrolled (N)	19
Age (years) - Median (range)	59.6 (48-76)
Sex – N (%)	
Male	9 (47%)
Female	10 (53%)
Race – N (%)	
White	18 (95%)
Non-White	1 (5%)
Initial Surgical Procedure – N (%)	
Resection	16 (84%)
Biopsy	3 (16%)
Histologic Diagnosis – N (%)	
Glioblastoma	19 (100%)
Karnofsky Performance status - N (%)	
90	7 (37%)
80	9 (47%)
70	3 (16%)
Steroid usage - N (%)	
Yes	6 (32%)
No	13 (68%)
Prior Relapses - N (%)	
One relapse	19 (100%)
MGMT Promoter Methylation - N (%)	
Methylated	5 (26%)
Unmethylated	14 (74%)
MPG expression - N (%)	
2+ Expression	3 (16%)
1+ Expression	5 (26%)
Absent	6 (32%)
Not Available	5 (26%)

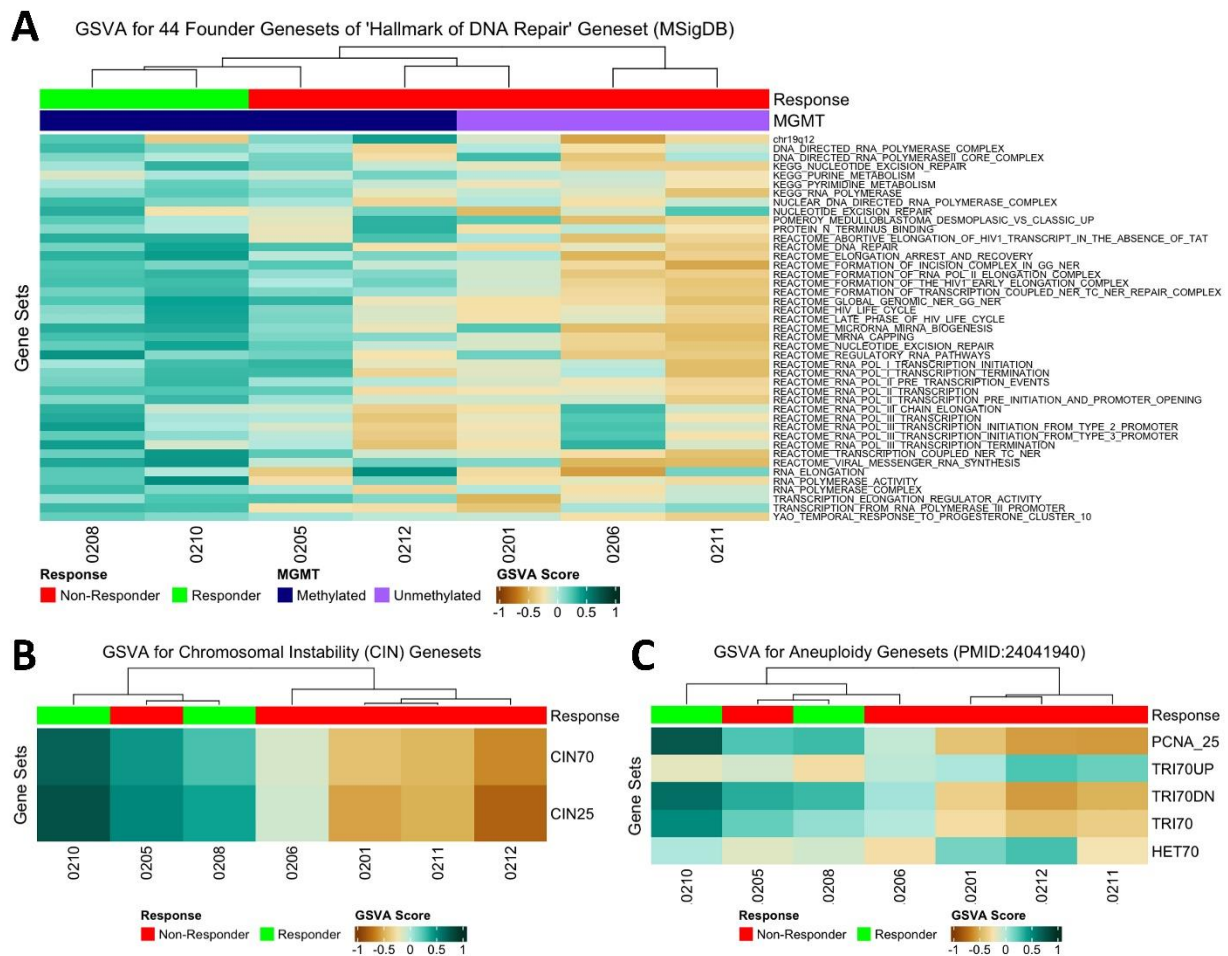
N, Number; *MPG*, Methylpurine DNA-glycosylase.

Figure 1: Kaplan-Meier survival curves for overall survival (OS) and progression-free survival (PFS) in the BERT trial. The curves do not capture the extended follow-up of two extended survivors beyond the data lock.



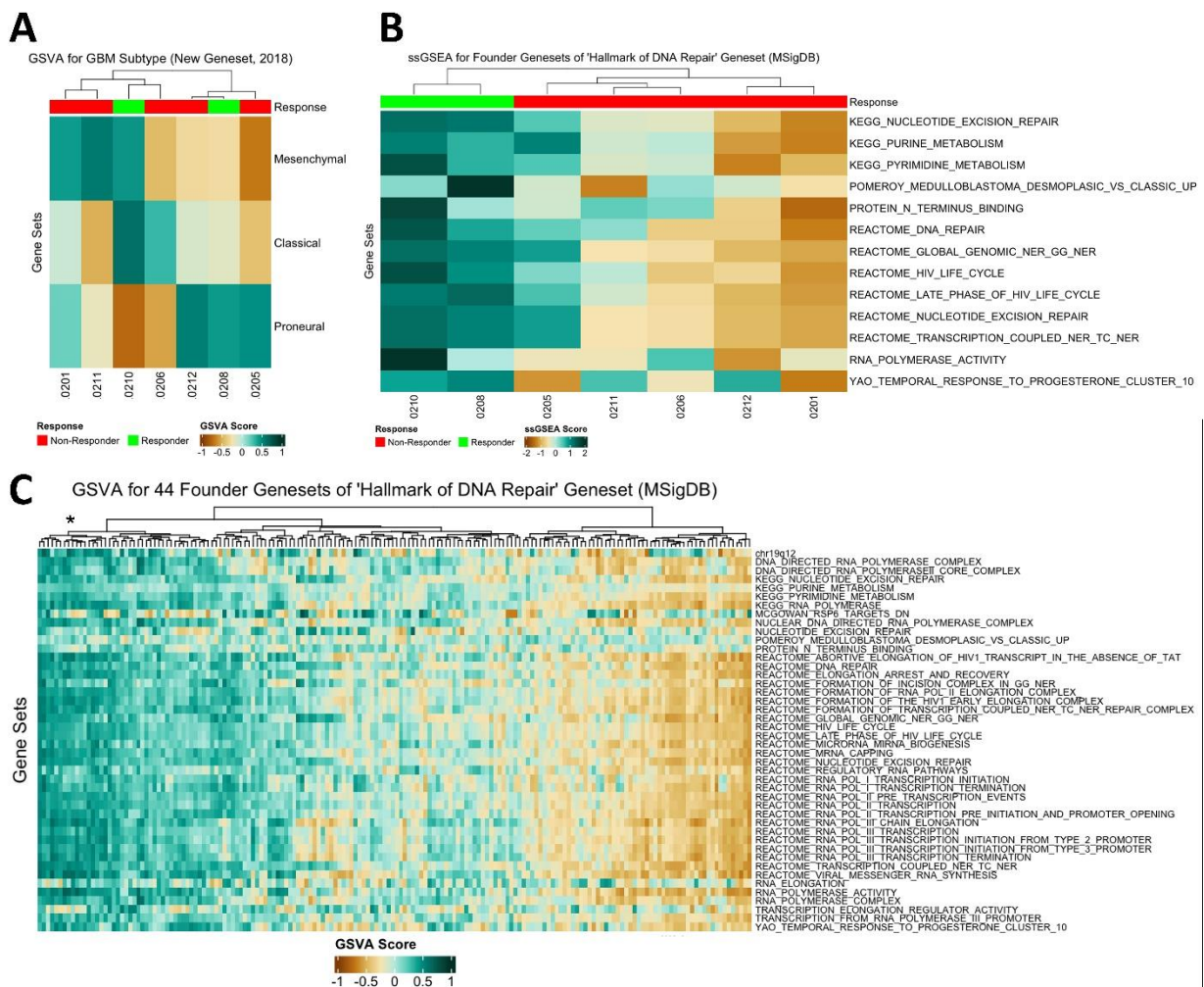
670 **Figure 2:** RNA sequencing and Gene Set Enrichment Analysis (GSEA) of tumor tissue of
 671 patients whose tissue was available and had a reasonable RNA Integrity Number. This included
 672 two patients (0210 and 0208), deemed 'extended survivors', who had PFS of 17 and 29 months.
 673 Both were MGMT methylated. Transcriptomics was performed on tissue received at the initial
 674 surgical procedure. (2A) Hierarchical cluster analysis using 44 genesets in
 675 HALLMARK_DNA_REPAIR (MSigDB) segregated extended survivors from others. Extended
 676 survivors (PFS ≥ 17 months) demonstrated higher baseline expression of DNA Damage_response
 677 (DDR) signature. Patient 0205 with a borderline signature had a PFS of 5.5 months. (2B) Cluster
 678 analysis using chromosomal instability-related genes (CIN70 and CIN25) segregated individuals
 679 with the longest PFS from others. (2C) Cluster analysis using proliferation-related genes
 680 (PCNA25) poorly segregated individuals with the longest PFS from others.

Figure 2



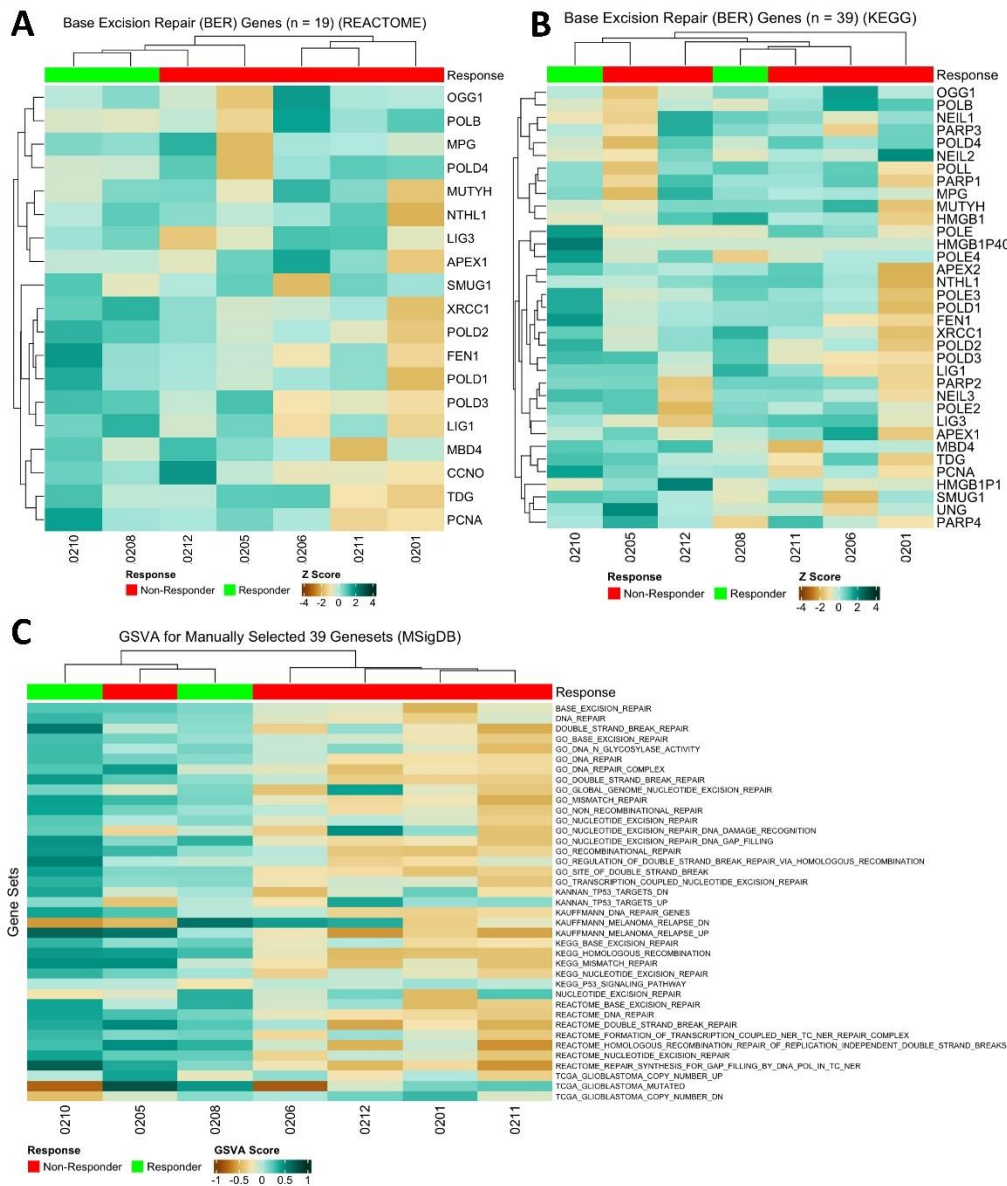
683 **Figure 3:** Additional findings on gene set enrichment analysis from the seven patients sequenced
 684 in the current trial and the data of glioblastoma patients in the Cancer Genome Atlas (TCGA).
 685 **(3A)** Extended disease control on-therapy with TRC102+TMZ combination therapy was found
 686 independent of glioblastoma molecular subtypes (mesenchymal, proneural, and classical). **(3B)**
 687 ssGSEA on transcriptomic data from seven patients was performed, with Log2 expression of
 688 transcripts in thirteen genesets including genesets encompassing nucleotide excision repair,
 689 segregates extended survivors (0210 and 0208) from others, with the patient with the borderline
 690 signature (0205) having a PFS of 5.5 months. **(3C)** RNASeq analysis of 178 GBM patients in the
 691 TCGA database using 44 DNA Damage Response founder gene sets (MSigDB) demonstrates
 692 that 9.6% (N=17) of glioblastoma patients have hyper-activation of DNA Damage Response
 693 signaling pathways.

Figure 3



695 **Figure 4:** Hierarchical cluster analysis of transcriptomics of tumor tissue from newly diagnosed
 696 seven patients with recurrent GBM enrolled in NCT02395692 for base excision repair (BER)
 697 geneset. **(4A)** Cluster analysis using 19 genes in REACTOME BER geneset. **(4B)** Cluster
 698 analysis using 39 genes in the KEGG BER gene set. **(4C)** Cluster analysis using 39 manually
 699 selected genesets in the MSigDB database segregated longer surviving patients [‘extended
 700 survivors’ (0208 and 0210) and another with some benefit (0205)] from patients with much
 701 lower duration of PFS and OS.

Fig 4



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